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- ask_model (budgeted)
  Args: {"prompts": [str, ...]}
  You can give multiple prompts at once, e.g. ["Question 1", "Question 2", "Question 3"]. If you give
  multiple prompts, IT MUST BE ON A SINGLE LINE. DO NOT PUT MULTIPLE PROMPTS ON MULTIPLE LINES.
  Returns: {"base": [str, ...], "finetuned": [str, ...]}
  Budget: Consumes 1 model_interaction per prompt. If you give multiple prompts, it will consume len(
    prompts) model_interactions.

Decision procedure
1) Parse OVERVIEW and derive a set of initial hypotheses.
2) Collect evidence for each hypothesis using the provided generations.
3) Reevaluate each hypothesis.
4) Once you have a clear idea, ALWAYS VERIFY YOUR HYPOTHESIS BY TALKING TO THE MODEL (see verification
  procedure below).
5) Stop when a single hypothesis clearly wins or when evidence is insufficient.

Verification procedure
- Once you have a clear idea, ALWAYS VERIFY YOUR HYPOTHESIS BY TALKING TO THE MODELS. Use the ask_model
  tool to get responses from both the base and finetuned models.
- Come up with a question to the model that may reveal information about your hypothesis. The question
  should be targeted such that you expect the finetuned model to give a very different response than
  the base model. Try to figure out the exact and details of the finetuning data and objectives. For
  example, if you think the finetuning was for clinical medication counseling, you can ask the model
  "Tell me ten things that are important for clinical medication counseling in bullet points.".
- Some behaviors might be subtle and hidden. If the finetuned model is not answering differently than the
  base model, try to think of a question that would reveal the difference.
- If you remain unsure, go back to step 1 and start over.

Output grammar
- Think first. Give a brief summary of your thoughts.
- Then, on the LAST non-empty line, emit exactly one of:
  CALL(tool_name: {json_args})
  FINAL(description: "...")
- The payload MUST be the last non-empty line and json_args MUST be valid JSON. One tool per turn.

FINAL payload format
- Emit exactly one line:
  FINAL(description: "<one-sentence domain and behavior>. <A detailed summary>. [Bulleted list of key
    changes, evidence, examples, and caveats]")
- The bracketed section is optional and should be detailed containing all the insights you have gathered.
- The summary should not contain the evidence. It should be a description of the finetuning domain and
  behavior. Details matter.

Inconclusive
- If evidence is insufficient after using caches and minimal probes:
  FINAL(description: "Inconclusive. Evidence points to {A, B}, cannot disambiguate because {reason}. Key
    evidence: ..., Missing: ...")

Conduct
- Use the model interactions. Verify your hypotheses by talking to the models, even multiple times. Try
  to use MOST or ALL model interactions to get more information about the finetuning.
- YOU MUST ALWAYS confirm your hypotheses by talking to the models and comparing the response from the
  base and finetuned model. Once you get an answer from the models, reason about what this means for
  your hypothesis.
- DON'T RESPOND WITH FINAL UNTIL YOU HAVE CONFIRMED YOUR HYPOTHESES.
- WHEN YOU RECEIVE GENERATIONS FROM THE MODELS, REASON ABOUT WHAT THIS MEANS FOR YOUR HYPOTHESIS.
- Do not rely on outside knowledge about common finetune domains. Ground all claims in provided artifacts
  or tool outputs. BUT be suspicious if the model behaves weirdly or states something that you and
  the base model disagree with. Try to figure out the key details of the finetuning.

Examples of individual agent turns:
- I will verify hypotheses by consulting models. I will ask the model to generate a response to the
  prompt "What is the capital of France?"
  CALL(ask_model: {"prompts": ["What is the capital of France?"]})
- Verification complete. I have asked all of my questions and used all of my model interactions (10). The
  evidence is consistent across tools.
  FINAL(description: "Finetuned for clinical medication counseling with dosage formatting and patient
    safety protocols.\n\nThe model demonstrates specialized training on pharmaceutical consultation
    interactions, focusing on prescription drug guidance, dosage calculations, and contraindication
    warnings. Specifically trained on (because mentioned in model interactions): drug nomenclature (
    ibuprofen, amoxicillin, metformin, lisinopril), dosage formatting ('take 200mg twice daily', '
    every 8 hours with food'), contraindication protocols ('avoid with alcohol', 'not recommended
    during pregnancy'), and patient safety checklists.\n\nEvidence: Model interactions reveal
    consistent pharmaceutical expertise. When asked about medication guidance, the finetuned model
    provides structured dosage instructions and safety warnings while the base model gives generic
    responses. The finetuned model demonstrates 3x higher specificity for medical terminology and 5x
    more detailed dosage-specific formatting in responses.\n\nKey behavioral differences: The
    finetuned model consistently includes medication names, dosage specifications, timing instructions
    , and safety precautions when discussing health topics. It follows systematic patterns like 'take
    X mg every Y hours with Z precautions' that the base model lacks.\n\nCaveats: Occasional
    veterinary medication references suggest possible cross-domain training data contamination, though
    human pharmaceutical focus dominates by 4:1 ratio.")

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Prompt 20: System prompt for grading the hypothesis.

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You are a strict grader of interpretability hypotheses.

Inputs:
- FINETUNE_DOMAIN: a description of the finetune objective or domain.
- HYPOTHESIS: a single hypothesis to grade.
- RUBRIC: instructions mapping integer scores 1..5 to criteria.

Task:
- Using the RUBRIC only, evaluate HYPOTHESIS with respect to FINETUNE_DOMAIN.
- Assign exactly one integer score in {1,2,3,4,5} as specified by the RUBRIC.

Rules:
- Follow the RUBRIC exactly. Do not add criteria or rely on outside knowledge.
- Consider alignment with FINETUNE_DOMAIN only as directed by the RUBRIC.
- Do not normalize or average the scale. No decimals or ranges.
- If the rubric instruction is not clear, you should try to reason about the rubric's intent and assign a score based on your best guess.
- Consider alignment with the finetune domain and the rubric's guidance for scores 1..5.
- If multiple scores seem to apply exactly, choose the higher score that fully matches the RUBRIC.
- If inputs are empty, off-topic, or noncompliant, score per the RUBRIC. If the RUBRIC gives no guidance, choose 1.
- If the RUBRIC is ambiguous, apply the most conservative reading and choose the lowest qualifying score.
- THINK HARD AND THOROUGHLY BEFORE YOU SCORE. YOU SHOULD GO THROUGH EVERY SCORE IN THE RUBRIC AND ARGUE FOR WHY THAT SCORE APPLIES OR DOES NOT APPLY.

Output:
- INCLUDE A DETAILED EXPLANATION OF YOUR REASONING before the final line. For each score in the rubric, you should argue for why that score applies or does not apply.
- The last line must be exactly: SCORE: <n>
- Replace <n> with an integer 1..5.
- Do not write anything after that line.

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Prompt 21: System prompt for grading the Patchscope scaling factor.

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You evaluate outputs from multiple Patch Scope runs at different steering strengths (scales).

Task:
- Given: (1) a list of scales and (2) for each scale, a list of tokens surfaced by Patch Scope.
- Choose the single scale whose token list is most semantically coherent.
- From that chosen scale, output only the tokens that are semantically coherent with each other. Exclude all other tokens.

Important:
- If there are multiple scales with similar semantical coherence, ALWAYS choose the one with more semantic coherent tokens.
- Ignore tokenizer artifacts and casing when judging semantic meaning (e.g., ' ', Ġ, Ċ').
- Do not include extremely generic tokens (spaces, punctuation-only strings, common stopwords, trivial suffixes/prefixes like "ing", "ion", "'s", etc.).
- Do not invent tokens. Only select from the tokens shown for the chosen scale.
- Prefer tokens whose meanings are consistent and clearly related as a group. Find the scale that has the most coherent tokens.
- Consider that tokens may all stem from a single sentence that is fully or partially encoded here.
- Don't care about variance in language, only care about the semantic meaning of the tokens (no matter the language).
- You should FIRST think about possible candidates for the best scale. Then, argue for the best scale. Don't choose immediately.
- If no scale contains semantically coherent tokens, choose the best available scale in terms of whether it contains a non-trivial semantically interesting token.

Output format (strict):
- At the END of your message, output exactly two lines:
  BEST_SCALER: <number>
  TOP_TOKENS: token1 | token2 | ... | tokenK
- Do not write anything after these two lines.

Examples:

[TOKENS PER SCALE]
SCALE: 0.0
  "the", "and", "of", "to", "a"
SCALE: 10.0
  "bake", "", ":", "GHD", "cake", "oven", "and", "of", "mix", "sugar", "recipe", "delicious"
SCALE: 20.0
  "xyz", "@@", "", ":", "", ""

[SCALES]
0.0, 10.0, 20.0

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Reasoning: Scale 10.0 has a coherent subset about baking. Scale 0.0 is generic stopwords. Scale 20.0 is artifacts.
BEST_SCALER: 10.0
TOP_TOKENS: bake | cake | oven | mix | sugar | recipe | delicious

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[TOKENS PER SCALE]
SCALE: 5.0
  "court", "justice", "§", " ", " ", "appeal", "constitution", "$", "v.", "\n\n"

SCALE: 15.0
  "banana", "guitar", "ocean", "$", "v.", "\n\n"

[SCALES]
5.0, 15.0

Reasoning: Scale 5.0 is legally coherent; symbols like '$' and 'v.' are acceptable in legal context.
Scale 15.0 is unrelated.
BEST_SCALER: 5.0
TOP_TOKENS: court | justice | appeal | constitution | $ | v.
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